**FIRST MODIFICATION,
JUNE 2025**

Appendix 9

Reclaim Cooling Towers

Operational Practices for Air Emissions Controls

Don Plant

Final

November 4, 2022

Table of Contents

2. Introduction

Simplot has developed Appendix 9: Reclaim Cooling Towers Operational Practices for Air Emissions Control to describe the Facility's current best operating practices including interim fluoride emissions reduction measures in place. Following the removal of the Reclaim Cooling Towers (or EPA approval of an Alternative Proposed Fluoride Reduction Plan under Paragraphs 32-33 of the Consent Decree), this Operational Practices for Air Emissions Control (Appendix 9) will no longer be in effect.

All capitalized terms and/or acronyms not otherwise defined in this Appendix shall have the meaning set forth in the Consent Decree.

2. Current Operating Practices

2.1 General

The Reclaim Cooling Towers are evaporative cooling towers. Evaporative cooling towers are open water recirculating devices that use fans or natural draft to draw or force ambient air through the device to remove heat from process water by direct contact.

Within the Phosphoric Acid Plant, the Reclaim Cooling Towers provide cooling for the phosphoric acid evaporators (except for Evaporator #12 – see Section 2.5) through a circulation of cooling water (including evaporator condensates) that directly contacts with process vapors in the evaporator barometric condensers. The process vapors add fluoride compounds to the circulating cooling water. Fluoride emissions from the Reclaim Cooling Towers are measured periodically. The testing requirements and fluoride emissions limits are documented in the Tier I Operation Permit No. T1-2017.0024 issued by the Idaho Department of Environmental Quality (IDEQ).

When the Reclaim Cooling circuit water level is low, make-up water is supplied to the cooling water supply basin. This make-up water is limited to either fresh well water or groundwater extraction well water.

2.2 Cooling Tower Fan Speeds

A major operational control feature of the cooling towers is the fan speed. The fans provide the air flow to provide the heat transfer from the condensates to ambient air. All of the fans of the Reclaim Cooling Towers are equipped with variable frequency drives that allow modulation of the fan speed. The Reclaim Cooling Tower fans are typically set at 90-100%. However, in the event a source (stack) test does not demonstrate compliance with the fluoride emission limit, the fan speed will be reduced to 85% and additional fluoride emission testing conducted.[1] The fan will then be operated at a reduced fan speed where compliance with the fluoride emission limit is demonstrated. The fan speed will not be increased until compliance with the fluoride emission limit can be demonstrated at that higher fan speed.

The following scenario demonstrates the procedures followed to set fan speed in the event a stack test does not demonstrate compliance.

[1] Historical operating data has shown that a fan speed of 85% results in a reduction of fluoride emissions.

1. Stack test on Cell “A” is conducted at normal (e.g., 90-100%) fan speed
    a. Results from the stack testing contractor (typically 2-6 weeks after sampling) does not demonstrate compliance with emission limit:
        i. Fan speed is reduced to 85% upon receipt of result;
        ii. Cooling tower cell is evaluated for any maintenance needs or other factors that might have contributed to a potential increase in emissions; and
        iii. Stack test scheduled to demonstrate compliance at reduced fan speed of 85%.
    b. If the stack test demonstrates compliance, then the fan speed will be maintained at 85%.

The following scenario outlines the procedures to be followed to increase fan speed after it has been reduced.

2. Stack test on Cell “A” is rescheduled at increased or normal (e.g. 90-100%) fan speed.
    a. Second stack test on Cell “A” conducted at increased or normal fan speed:
        i. Fan speed increased on the day of the test;
        ii. Retest conducted at increased fan speed; and
        iii. Fan speed returned to the reduced operating speed (typically 85%) immediately after completing the test.
    b. If the retest at increased or normal (e.g. 90-100%) fan speed demonstrates compliance with emission limit, then:
        i. After submittal of retest results to DEQ, fan speed is set at the normal or operating speed which is equal to, or lower than the successful retest.
    c. If the retest at increased or normal fan speed does not demonstrate compliance with emission limit, then:
        i. Fan speed remains at reduced speed (normally 85%).

2.3 Cooling Tower Operations & Scrubber Blowdown Routing

Scrubber blowdown streams are combined with the Process Wastewater from the Phosphoric Acid Plant, which (together) goes to the phosphogypsum stack. Decant process wastewater from the phosphogypsum stack is recycled back into the phosphoric acid process. A portion of the fluoride from the scrubber blowdown that goes to the Phosphogypsum Stack is routed to the Reclaim Cooling Towers only after being recycled back into the production process.

2.3.1 Typical Operation

The scrubbers that contribute scrubber blowdown to the Process Wastewater are the phosphoric acid digester scrubber, phosphoric acid belt filter scrubber, phosphoric acid tank farm scrubber, super phosphoric acid primary control scrubber, fluorine scrubber, super phosphoric acid extended absorber system, animal feed deflo scrubber, and animal feed Entoleter scrubber. The phosphoric acid and super phosphoric acid scrubber blowdowns flow into the blend system before being utilized to either fluidize the phosphogypsum going to the Phosphogypsum Stack System or as filter wash water on the phosphoric acid belt filters. The filter wash water is then recycled back to the phosphoric acid digester and incorporated into the phosphoric acid being produced. The animal feed scrubber blowdowns are used to fluidize the phosphogypsum going to the Phosphogypsum Stack System.

2.3.2 Evaporator Wash Operation

Another process stream (except for Evaporator #12 – see Section 2.5) that currently goes to the Reclaim Cooling Towers is evaporator wash water. Prior to completion of the compliance projects set forth in Appendix 6 to the Consent Decree, evaporator wash water is sent from the evaporator barometric condensers to the Reclaim Cooling Towers. The evaporator wash water is supplied by the #1 Wash Water Tank, which is an existing portion of the Acid Value Recovery System and currently receives inputs shown in Diagram 2 of the Facility Report.[2] Following completion of the compliance projects set forth in Appendix 6 to the Consent Decree, evaporator wash water will be supplied and recirculated back to the Acid Value Recovery System.

[2] Prior to completion of the compliance projects set forth in Appendix 6 to the Consent Decree, Diagram 2 of the Facility Report is not an accurate depiction of the effluent destinations of the existing portions of the Acid Value Recovery System. See Section 1.4 of the Consolidated Materials Management Practices document for further details.

2.4 Blend System

Simplot implemented the blend system in October 2016 to reduce the fluoride emissions from the Reclaim Cooling Towers. The blend system removed the phosphoric acid digester flash coolers from the Reclaim Cooling Tower circuit and re-directed this cooling and fluoride load directly to the Phosphogypsum Stack System.

The blend system enables a reduction in fluoride emissions in several ways. First, with the reduced heat load to the Reclaim Cooling Towers, the facility is currently targeting an annual average of two cooling tower cell fans idled.[3] Second, by reducing the heat load to the Reclaim Cooling Towers, the temperature of the water entering the cooling system is reduced, thus lowering the vapor pressures (and emissions) of volatile fluoride compounds. Furthermore, some of the fluoride in the flash cooler streams (that are sent to the gypsum stack) reacts with or is absorbed into the calcium sulfate (gypsum) crystal structure rather than being emitted into the atmosphere.[4] Finally, the blend system also reduces the amount of potential scrubber blowdown water flowing to the Reclaim Cooling Towers.

2.5 Evaporator #12 Cooling

Simplot will implement in May/June 2025 a change in the cooling for phosphoric acid Evaporator #12, which will further reduce the fluoride load to the Reclaim Cooling Towers (RTC). This modification removes Evaporator #12 from the RTC circuit and re-directs this heat and fluoride load directly to the Phosphogypsum Stack System.

The change in Evaporator #12 cooling method enables a potential reduction in fluoride emissions by: reducing the heat load to the RCT and sending the associated fluorides to the gypsum stack where a portion of the fluorides will react with or be absorbed into the calcium sulfate (gypsum) crystal.

[3] Utilizing the motor run status tags from the facility's process control system (i.e., the Distributive Control System - DCS) averaged over a day (midnight to midnight), the number of fans idled each day is calculated and is then averaged for the calendar year. The data collected so far shows a reduction of cooling tower cell operation of 1-1.3 "cells" annually.

[4] Precipitation reactions in the gypsum pond system include sodium and potassium fluosilicates, fluoroaluminates, chukrovite, and calcium fluoride. The net result of complexation and precipitation is a reduction in free fluoride concentration, which is the primary source of fluoride emissions.

[redacted]